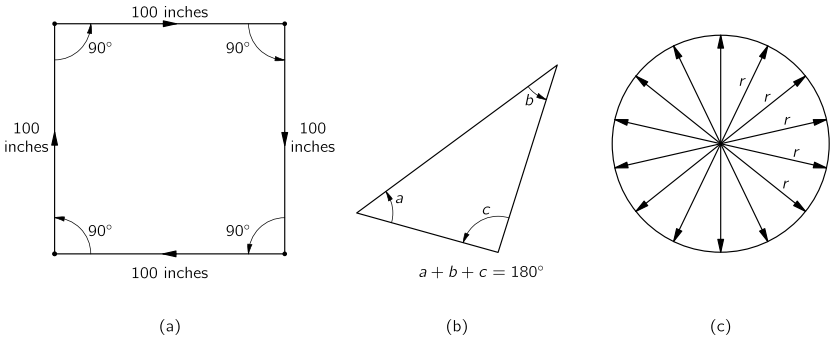

Figure 6-7 A square, triangle, and circle in a flat space.

a single point, and lay out a lot of dots that are all the same distance from that point. See Figure 6-7(c). (We have to be careful how we define these things because we've got to be able to make the analogs for the other fellows.) Of course, it's equivalent to the curve you can make by swinging a ruler around a point. Anyway, our bug learns how to make circles. Then one day he thinks of measuring the distance around a circle. He measures several circles and finds a neat relationship: The distance around is always the same number times the radius *r* (which is, of course, the distance from the center out to the curve). The circumference and the radius always have the same ratio—approximately 6.283—independent of the size of the circle.

Now let's see what our other bugs have been finding out about *their* geometries. First, what happens to the bug on the sphere when he tries to make a "square"? If he follows the prescription we gave above, he would probably think that the result was hardly worth the trouble. He gets a figure like the one shown in Figure 6-8. His end-point *B* isn't on top of the starting point *A*. It doesn't work out to a closed figure at all. Get a sphere and try it. A similar thing would happen to our friend on the hot plate. If he lays out four straight lines of equal length—as measured with his expanding rulers—joined by right angles he gets a picture like the one in Figure 6-9.